

PRACTICE SHEET

1. What is the HCF of polynomials $x^3+3x^2y+2xy^2$ and $x^4+6x^3y+8x^2y^2$?
 (1) $x(x+2y)$ (2) $x(x+3y)$
 (3) $(x+2y)$ (4) None of these
2. If the HCF of x^3+mx^2-x+2m and x^2+mx-2 is a linear polynomial, then what is the value of m ?
 (1) 1 (2) 2
 (3) 3 (4) 4
3. What is the HCF of $(x^2+bx-x-b)$ and $x^2+x(a-1)-a$?
 (1) $x+b(b)$ (2) $x+a$
 (3) $x+1(d)$ (4) $x-1$
4. If the HCF of x^2+x-12 and $2x^2-kx-9$ is $x-k$, then what is the value of k ?
 (1) -3 (2) 3
 (3) -4 (4) 4
5. What is the LCM of $(6x^3 + 60x^2+150x)$ and $(3x^4+12x^3-15x^2)$?
 (1) $6x^2(x+5)^2(x-1)$ (2) $3x^2(x+5)^2(x-1)$
 (3) $6x^2(x+5)^2(x-1)^2$ (4) $3x^2(x+5)(x-1)^2$
6. What is the LCM of $(x+2)^2(x-2)$ and $x^2-4x-12$?
 (1) $(x+2)(x-2)$
 (2) $(x+2)^3(x-2)(x-6)$
 (3) $(x+2)(x-2)^2$
 (4) $(x+2)2(x-2)(x-6)$
7. What is the value of k for which the HCF of $2x^2+kx-12$ and $x^2+x-2k-2$ is $(x+4)$?
 (1) 5 (2) 7
 (3) 10 (4) -4
8. What is the LCM of $(x^2-y^2-z^2-2yz)$, $(x^2-y^2+z^2+2xz)$ and $(x^2+y^2-z^2-2xy)$?
 (a) $(x+y+z)(x+y-z)(x-y+z)$
 (b) $(x+y+z)(x-y-z)(x-y+z)$
 (c) $(x+y+z)(x+y-z)(x-y-z)$
 (d) $(x+y-z)(x-y-z)(x-y+z)$
9. If $(x+2)$ is the HCF of x^2+ax+b and x^2+cx+d then which one of the following is correct?
 (1) $a+c=b+d$ (2) $2a+b=2c+d$
 (3) $b+2c=2a+d$ (4) $b-2c=2a-d$
10. What are the values of c when the HCF of x^3+cx^2-x+2c and x^2+cx-2 over the rational is a linear polynomial?
 (1) ± 1 (2) ± 2
 (3) ± 3 (4) ± 4
11. If the HCF of x^3-27 and $x^3+4x^2+12x+k$ is a quadratic polynomial, then what is the value of k ?
 (1) 27 (2) 9
 (3) 3 (d) -3
12. What is the LCM of $3(a^3-b^3)$ and $11(a^4-b^4)$?
 (1) $33(a^3-b^3)(a^2+b^2)(a+b)$
 (2) $(a^3-b^3)(a^2+b^2)(a-b)$
 (3) $33(a-b)(a^2+b^2-ab)(a^3-b^3)$
 (4) $33(a^3-b^3)(a^4-b^4)$
13. I. The HCF of $x+y$ and $x^{10}-y^{10}$ is $x+y$.
 II. The HCF of $x+y$ and $x^{10}+y^{10}$ is $x+y$.
 III. The HCF of $x-y$ and $x^{10}+y^{10}$ is $x-y$.
 IV. The HCF of $x-y$ and $x^{10}-y^{10}$ is $x-y$.
 Which of the statement given above are correct?
 (1) I and II (2) II and III
 (3) I and IV (4) II and IV
14. If $f(x)$ and $g(x)$ are two polynomials with integral co-efficient which vanish at $x=1/2$, then what is the factor of HCF of $f(x)$ and $g(x)$?
 (1) $x-1$ (2) $x-2$
 (3) $2x-1$ (4) $2x+1$
15. What is the HCF of $4x^3+3x^2y-9xy^2+2y^3$ and $x^2+xy-2y^2$?
 (1) $x-2y$ (2) $x-y$
 (3) $(x+2y)(x-y)$ (4) $(x-2y)(x-y)$
16. The HCF of two polynomials $p(x)$ and $q(x)$ is $2x(x+2)$ and LCM is $24x(x+2)^2(x-2)$. If $p(x)=8x^3+32x^2+32x$, then what is $q(x)$ equal to?
 (1) $4x^3-16x$ (2) $6x^3-24x$
 (3) $12x^3+24x$ (4) $12x^3-24x$

17. What is the HCF of $(x^4 - x^2 - 6)$ and $(x^4 - 4x^2 + 3)$?
 (1) $x^2 - 3$ (2) $x + 2$
 (3) $x + 3$ (4) $x^2 + 3$
18. What is the LCM of $a^3b - ab^3$, $a^3b^2 + a^2b^3$ and $ab(a + b)$?
 (1) $a^2b^2(a^2 - b^2)$ (2) $ab(a^2 - b^2)$
 (3) $a^2b^2 + ab^3$ (4) $a^3b^3(a^2 - b^2)$
19. What is the HCF of the polynomials $x^4 - 3x + 2$, $x^3 - 3x^2 + 3x - 1$ and $x^4 - 1$?
 (1) $x - 1$ (2) $x + 1$
 (3) $x^2 - 1$ (4) None of these
20. What is the HCF of $(x^2 - x - 6)$ and $(x^2 - 4x + 3)$?
 (1) $x^2 - 3$ (2) $x + 2$
 (3) $x - 3$ (4) $x^2 + 3$

ANSWER KEY

1.	(1)	2.	(1)	3.	(4)	4.	(2)	5.	(1)	6.	(4)	7.	(1)	8.	(2)	9.	(3)	10.	(1)
11.	(2)	12.	(1)	13.	(3)	14.	(3)	15.	(3)	16.	(2)	17.	(1)	18.	(1)	19.	(1)	20.	(3)

CDS PYQ

1. What is the HCF of the polynomials x^3+8 , x^2+5x+6 and x^3+2x^2+4x+8 ?
[CDS 2013(I)]
 (a) $x+2$ (b) $x+3$
 (c) $(x+2)^2$ (d) None of these
2. The LCM of (x^3-x^2-2x) and (x^3+x^2) is
[CDS 2013(I)]
 (a) x^3-x^2-2x (b) x^2+x
 (c) $x^4-x^3-2x^2$ (d) $x-2$
3. The HCF of (x^4-y^4) and (x^6-y^6) is
[CDS 2013(I)]
 (a) x^2-y^2 (b) $x-y$
 (c) x^3-y^3 (d) x^4-y^4
4. What is the LCM of x^2+2x-8 , x^3-4x^2+4x and x^2+4x ?
[CDS 2013(II)]
 (a) $x(x+4)(x-2)^2$ (b) $x(x+4)(x-2)$
 (c) $x(x+4)(x+2)^2$ (d) $x(x+4)^2(x-2)$
5. What is the HCF of $a^2b^4 + 2a^2b^2$ and $(ab)^7-4a^2b^9$?
[CDS 2013(II)]
 (a) ab (b) a^2b^3
 (c) a^2b^2 (d) a^3b^3
6. What is the HCF of $8(x^5-x^3+x)$ and $28(x^6+1)$?
[CDS 2014(I)]
 (a) $4(x^4-x^2+1)$ (b) $2(x^4-x^2+1)$
 (c) (x^4-x^2+1) (d) None of these
7. What is the lowest common multiple of $ab(x^2 + 1) + x(a^2 + b^2)$ and $ab(x^2 - 1) + x(a^2 - b^2)$?
[CDS 2014 (2)]
 (a) $(a^2x^2 - b^2)(a + bx)$
 (b) $(a^2x^2 - b^2)(a + bx)^2$
 (c) $(a^2x^2 - b^2)(a - bx)$
 (d) $(a^2x^2 - b^2)(a - bx)^2$
8. What is the highest common factor of $2x^3 + x^2 - x - 2$ and $3x^3 - 2x^2 + x - 2$?
[CDS 2014 (2)]
 (a) $x-1$ (b) $x + 1$
 (c) $2x+1$ (d) $2x-1$
9. The HCF and LCM of two polynomials are $(x+y)$ and $(3x^5+5x^4y+2x^3y^2-3x^2y^3-5xy^4-2y^5)$ respectively. If one root of the polynomial is (x^2-y^2) , then the other polynomial is
[CDS 2015(I)]
 (a) $3x^4-8x^3y+10x^2y^2+7xy^3-2y^4$
 (b) $3x^4-8x^3y-10x^2y^2+7xy^3+2y^4$
 (c) $3x^4+8x^3y+10x^2y^2+7xy^3+2y^4$
 (d) $3x^4+8x^3y-10x^2y^2+7xy^3+2y^4$
10. If $(x+1)$ is the HCF of Ax^2+Bx+C and Bx^2+Ax+C where $A \neq B$, then the value of C is
[CDS 2015(II)]
 (a) A (b) B
 (c) $A-B$ (d) 0
11. The sum and difference of two expressions are $5x^2-x-4$ and $x^2+9x-10$ respectively. The HCF of the two expression will be
[CDS 2016(I)]
 (a) $(x+1)$ (b) $(x-1)$
 (c) $(3x+7)$ (d) $(2x-3)$

12. What is the LCM of x^3+8 , x^2+5x+6 and x^3+4x^2+4x ?

[CDS 2017(I)]

- (a) $x(x+2)^2(x+3)(x^2-2x+4)$
(b) $x(x-2)^2(x-3)(x^2+2x+4)$
(c) $(x+2)^2(x+3)(x^2-2x+4)$
(d) $(x-2)^2(x-3)(x^2-2x+4)$

13. The product of two non zero expressions is $(x+y+z)p^3$. If their LCM is

[CDS 2017(II)]

- (a) $(x+y+z)$ (b) $(x+y+z)p^2$
(c) $(x+y+z)p^5$ (d) $(x+y+z)p$

14. If the HCF of polynomials $f(x) = (x-1)(x^2+3x+a)$ and $g(x) = (x+2)(x^2+2x+b)$ is (x^2+x-2) , then what are the values of a and b respectively?

[CDS 2017(II)]

- (a) 2, 2 (b) 2, -3
(c) -1, -3 (d) -2, -1

15. HCF and LCM of two polynomials are $(x+3)$ and $(x^3-9x^2-x+105)$ respectively. If one of the two polynomials is $x^2-4x-21$, then the other is

[CDS 2018(II)]

- (a) $x^2-2x-21$ (b) $x^2+2x+15$
(c) $x^2-2x-15$ (d) x^2-x-15

16. What is the LCM of the polynomials x^3+3x^2+3x+1 , x^3+5x^2+5x+4 and x^2+5x+4 ?

[CDS 2019(II)]

- (a) $(x+1)^3(x+4)(x^2+x+1)$
(b) $(x+4)(x^2+x+1)$
(c) $(x+1)(x^2+x+1)$
(d) $(x+1)^2(x+4)(x^2+x+1)$

17. If $(x+k)$ is the HCF of x^2+5x+6 and $x^2+8x+15$, then what is the value of k

[CDS 2020(I)]

- (a) 5 (b) 3
(c) 2 (d) 1

18. What is the HCF of the polynomials $x^6-3x^4+3x^2-1$ and x^3+3x^2+3x+1

[CDS 2020(II)]

- (a) $(x+1)$ (b) $(x+1)^2$
(c) x^2+1 (d) $(x+1)^3$

19. HCF and LCM of two polynomials are $(3x+1)$ and $(30x^3+7x^2-10x-3)$ respectively. If one of the two polynomials is $6x^2+5x+1$, then the other is

[CDS 2020(II)]

- (a) $15x^2+4x+3$ (b) $15x^2+4x-3$

- (c) $15x^2-4x+3$ (d) $15x^2-4x-3$

20. If $(x-k)$ is the HCF of x^2+ax+b and x^2+cx+d , then what is the value of k?

[CDS 2021(I)]

- (a) $\frac{d-b}{c-a}$ (b) $\frac{d-b}{a-c}$
(c) $\frac{d+b}{c+a}$ (d) $\frac{d-b}{c+a}$

21. What is the HCF of $x^3-19x+30$ and x^2-5x+6 ?

[CDS 2021(II)]

- (a) $(x+2)(x-3)$ (b) $(x-2)(x+3)$
(c) $(x+2)(x-1)$ (d) $(x-2)(x-3)$

22. What is the HCF of $(x^8 - y^8)$ and $(x^7 - y^7 + x^5y^2 - x^2y^5)$?

[CDS 2022 (II)]

- (a) $(x^2 + y^2)$
(b) $(x^2 - y^2)$
(c) $(x^3 - y^3 - x^2y + xy^2)$
(d) $(x^3 - y^3 + x^2y - xy^2)$

23. If $(x + k)$ is the HCF of $x^2 + px + q$ and $x^2 + qx + p$, where $p \neq q$, then what is the value of k?

[CDS 2022 (II)]

- (a) -1 (b) 0
(c) $\frac{1}{2}$ (d) 1

24. Let $p(x) = x^4 + x^2 + 1$, $q(x) = x^4 - 2x^3 + 3x^2 - 2x + 1$. If GCD of $p(x)$ and $q(x)$ is $x^2 - x + 1$, then what is their LCM?

[CDS 2022 (II)]

- (a) $(x^2 + x + 1)(x^2 - x + 1)^2$
(b) $(x^4 + x^2 + 1)^2(x^2 - x + 1)$
(c) $(x^4 + x^2 + 1)(x^2 + x + 1)^2$
(d) $(x^4 + x^2 + 1)(x^2 - x + 1)^2$

25. What is the HFC of $acx^3 + bcx^2 + adx^2 + acdx + bdx + bcd$ and $adx^3 + acx^3 + bdx^2 + bcx + acdx + bcd$ if HCF $(c, d)=1$, $c \neq d$?

[CDS-2023 (2)]

- (a) $bx + c$ (b) $cx + d$
(d) $ax + d$ (d) $ax + b$

26. What is the LCM of $x^4 + x^2y^2 + y^4$, $x^3 + y^3$, $x^3 - y^3$?

[CDS-2023 (2)]

- (a) $(x^2 - y^2)(x^4 + x^2y^2 + y^4)^2$
(b) $(x^2 - y^2)(x^4 + 2x^2y^2 + y^4)$
(c) $(x^6 - y^6)$
(d) $(x^6 - y^6)$

27. What is the HCF of $x^4 - 13x^2y^2 - 300y^4$, $x^3 - 4x^2y - 4xy^2 - 5y^3$ and $x^3 - 125y^3$?

(a) $x - 5y$ (b) $x + 5y$ [CDS-2024 (1)] (c) $x^2 + 5xy + 25y^2$ (d) 1

ANSWER KEY

1.	(a)	2.	(c)	3.	(a)	4.	(a)	5.	(c)	6.	(a)	7.	(a)	8.	(a)	9.	(c)	10.	(d)
11.	(b)	12.	(a)	13.	(b)	14.	(b)	15.	(c)	16.	(a)	17.	(b)	18.	(d)	19.	(d)	20.	(b)
21.	(d)	22.	(a)	23.	(a)	24.	(a)	25.	d	26.	(c)	27.	(a)						